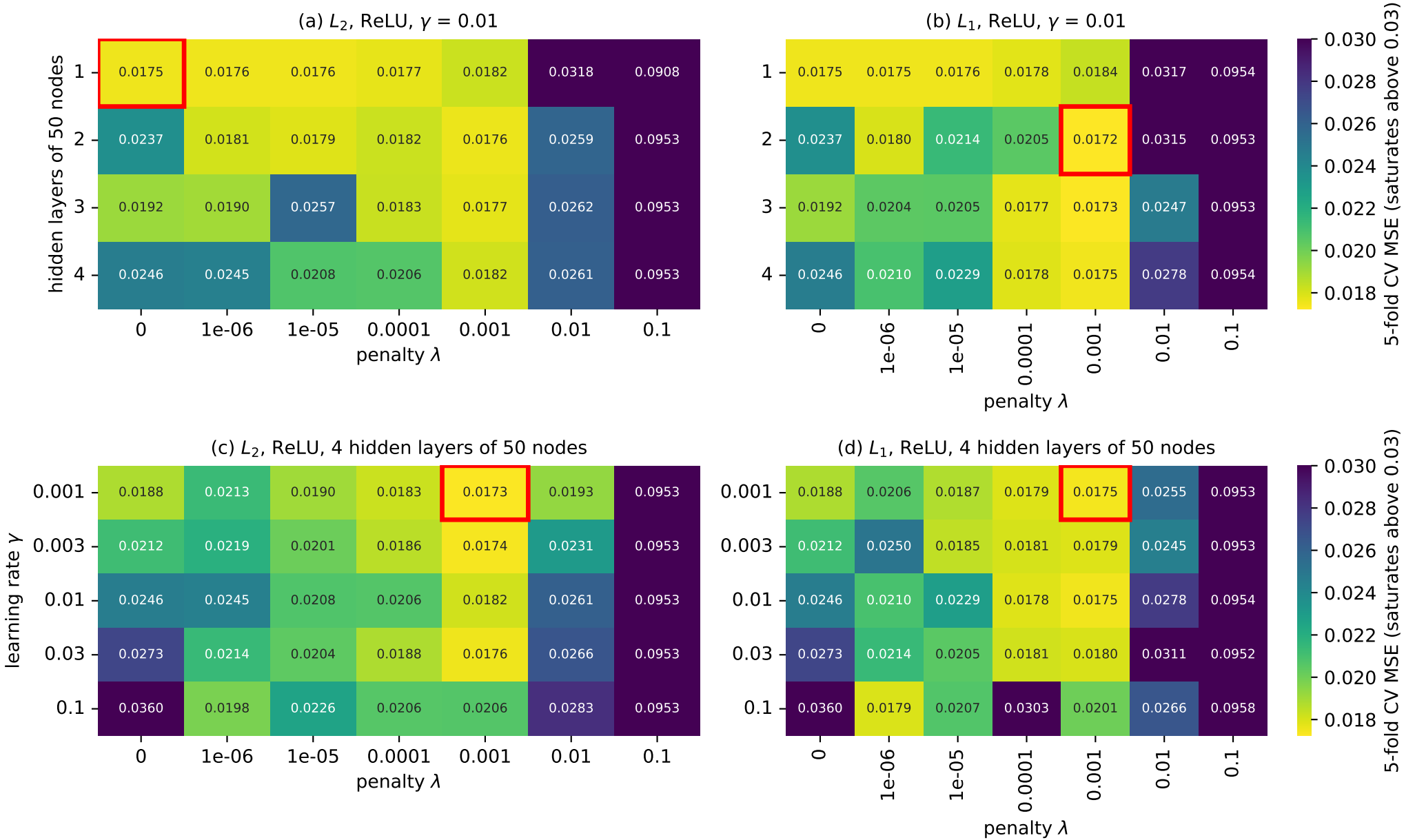
(a) L_2, ReLU, γ = 0.01
hidden layers of 50 nodes
1
2
3
4
0.0175 0.0176 0.0176 0.0177 0.0182 0.0318 0.0908
0.0237 0.0181 0.0179 0.0182 0.0176 0.0259 0.0953
0.0192 0.0190 0.0257 0.0183 0.0177 0.0262 0.0953
0.0246 0.0245 0.0208 0.0206 0.0182 0.0261 0.0953
0
1e-06
1e-05
0.0001
0.001
0.01
0.1
penalty λ
(b) L_1, ReLU, γ = 0.01
1
2
3
4
0.0175 0.0175 0.0176 0.0178 0.0184 0.0317 0.0954
0.0237 0.0180 0.0214 0.0205 0.0172 0.0315 0.0953
0.0192 0.0204 0.0205 0.0177 0.0173 0.0247 0.0953
0.0246 0.0210 0.0229 0.0178 0.0175 0.0278 0.0954
0
1e-06
1e-05
0.0001
0.001
0.01
0.1
penalty λ
0.030
0.028
0.026
0.024
0.022
0.020
0.018
5-fold CV MSE (saturates above 0.03)
(c) L_2, ReLU, 4 hidden layers of 50 nodes
learning rate γ
0.001
0.003
0.01
0.03
0.1
0.0188 0.0213 0.0190 0.0183 0.0173 0.0193 0.0953
0.0212 0.0219 0.0201 0.0186 0.0174 0.0231 0.0953
0.0246 0.0245 0.0208 0.0206 0.0182 0.0261 0.0953
0.0273 0.0214 0.0204 0.0188 0.0176 0.0266 0.0953
0.0360 0.0198 0.0226 0.0206 0.0206 0.0283 0.0953
0
1e-06
1e-05
0.0001
0.001
0.01
0.1
penalty λ
(d) L_1, ReLU, 4 hidden layers of 50 nodes
0.001
0.003
0.01
0.03
0.1
0.0188 0.0206 0.0187 0.0179 0.0175 0.0255 0.0953
0.0212 0.0250 0.0185 0.0181 0.0179 0.0245 0.0953
0.0246 0.0210 0.0229 0.0178 0.0175 0.0278 0.0954
0.0273 0.0214 0.0205 0.0181 0.0180 0.0311 0.0952
0.0360 0.0179 0.0207 0.0303 0.0201 0.0266 0.0958
0
1e-06
1e-05
0.0001
0.001
0.01
0.1
penalty λ
0.030
0.028
0.026
0.024
0.022
0.020
0.018
5-fold CV MSE (saturates above 0.03)